



Replacement of semi-natural cover with artificial substrates in urban parks causes a decline of house sparrows *Passer domesticus* in Mediterranean towns

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Abstract

Many European towns and cities have undergone reurbanisation processes in recent decades. However, their effects on urban biodiversity have been poorly studied. Currently urbanisation processes include park remodelling, which involves the replacement of semi-natural substrates (natural grass, bare soil) with artificial ones (pavement, concrete, areas for dogs, artificial grass). Our aim was to explore, for the first time, the effects of park remodelling on the trends and abundance of a declining urban exploiter, the House Sparrow. An abundance index of House Sparrows was obtained in 32 urban parks of four towns in the Valencian Community (Spain) in four summers (2015–2018). Of the studied parks, 10 were remodelled during the study period. Before remodelling, the trends and abundance of House Sparrows in both groups of parks were similar and stable on average, which suggest no bias between both park groups. However, House Sparrow abundance was significantly reduced in parks affected by remodelling works while in those non-remodelled it remained stable. Park remodelling might be linked to a reduction in both habitat suitability and availability of trophic resources, which could harm urban House Sparrows populations and possibly other species as well. Therefore, new park policies and urban planning measures are urgently needed to preserve urban House Sparrows.

Keywords Artificial grass · Park remodelling · Paving · Concrete · Areas for dogs · Reurbanisation

Introduction

Urban areas are expanding across the globe and their growth has increased exponentially during recent decades, usually at the expense of rural or suburban areas, and is one of the most important threats to biodiversity in the world (Isaksson 2018). The urbanisation of natural landscapes can affect wildlife through the elimination of some species and the entry of new ones that accept the proximity of humans and the new challenges that this implies, such as chemical pollution, new food resources, new predators or acoustic pollution (Herrera-Dueñas et al. 2017; Isaksson 2018; Shryock et al. 2017). Once

urbanisation has been well-established, it is not unusual to change the landscape again through a process that implies new urban organisation, defined as “reurbanisation” or “urban renewal” (Bouzarovski et al. 2010; Buzar et al. 2007a; Van den Berg et al. 1982; Vermeijden 2001). At the end of the twentieth century and at the beginning of the present century, most European cities and towns have undergone reurbanisation, which consists in the diversification of household structures, housing functions and the repopulation of inner-city neighbourhoods (Buzar et al. 2007b; Cheshire 2006; Vinuesa 2005), and is also linked to higher human socio-economic levels (Shaw et al. 2008).

Reurbanisation does not impact natural habitats, but causes degradation, modification or loss of existing urban green spaces, such as urban parks and/or private gardens, which are the main habitat of urban wildlife (Bernat-Ponce et al. 2018; Chamberlain et al. 2009; Cornelis and Hermy 2004; De Coster et al. 2015; Peris and Montelongo 2014; Verbeeck et al. 2011). These green urban spaces are especially important for birds during breeding seasons as they constitute the most important source of invertebrate food, which is essential for their breeding success (Chamberlain et al. 2007, 2009; Peach

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et al. 2015; Vincent 2005). Current new changes in urban green landscapes, linked to reurbanisation, together with increasing human socio-economic status, tend to remodel parks to meet human purposes which make these habitats less suitable for urban biodiversity. Remodelling activities potentially affecting biodiversity include: the paving of green areas (Mitschke and Mulsow 2003), increasing use of pesticides (Summers-Smith 2003), using sprinklers instead of hoses to water lawns (Murgui and Macias 2010), or building areas for dogs, paving children's play areas with rubber, and replacing natural lawn with artificial grass. These park remodelling actions linked to reurbanisation processes are transforming traditional parks into domestic modified versions of them. For example, replacing natural lawn with artificial grass is seen as a way to save water (Moore 2009) and to reduce management requirements in Mediterranean climates. These changes could lead to a significant reduction of the diversity and number of available invertebrates which could have an important effect limiting the reproductive success and survival of urban bird species (Chamberlain et al. 2009; Peach et al. 2015).

The House Sparrow *Passer domesticus* has been, and still remains, the most closely associated wild bird species with humans since prehistory (Anderson 2006; Ericson et al. 1997). However, throughout the twentieth century and at the beginning of the present one, the population of this human-adapted species has drastically declined in urban and rural areas (Hayhow et al. 2014; Summers-Smith 2003). In London, its decline was first noticed in the 1920s when engines replaced the horses that pulled trams (De Laet and Summers-Smith 2007; Summers-Smith 2005). Almost 50 years ago, House Sparrow populations in large cities of Western Europe began showing a new declining trend (Hayhow et al. 2014; Summers-Smith 2005, 2007).

The causes of urban House Sparrow decline still seem unclear and researchers have proposed several potential drivers, including: increased urban pollution (Herrera-Dueñas et al. 2017; Peach et al. 2018; Summers-Smith 2007); lack of nest sites due to the structure of the new buildings (Moudrá et al. 2018); an increased human socio-economic status (Shaw et al. 2008); poor quality of available trophic resources (Herrera-Dueñas et al. 2015); limited access to household waste after replacing communal surface trash containers with underground ones (Bernat-Ponce et al. 2018); increased electromagnetic radiation (Balmori and Hallberg 2007); increased predation by Eurasian Sparrow hawks *Accipiter nisus* or domestic cats *Felis silvestris catus* (Bell et al. 2010; Thomas et al. 2012) and increased exposure to avian malaria (Dadam et al. 2019).

Bird species are likely to present negative population trends in the most modified environments (Siriwardena et al. 1998), and even though urban House Sparrows have adapted to human presence and are considered urban exploiters (Isaksson 2018), they seem less abundant in highly urbanized zones with artificial surfaces (roads, streets, pavement networks or

parking spaces), which could be linked to this reurbanisation concept. Furthermore, remodelling their main feeding habitats, such as urban parks and green areas, could have strong effects on their populations (Bernat-Ponce et al. 2018; Murgui 2009; Šálek et al. 2015; Shaw et al. 2008). House Sparrows need green stepping stones to maintain an exchange between key urban populations (De Laet and Trappeniers 2019). The reduced suitability of remodelled urban parks could mean the loss of green stepping stones which could, in turn, have negative impacts on populations due to the limited home ranges of this species in urban areas (Heij 1985).

Here, our aim was to explore if park remodelling (including building areas for dogs, paving children's play areas with rubber, paving with concrete and asphalt, and/or replacing lawn with artificial grass), which is linked to the reurbanisation of towns and cities, negatively affects House Sparrow populations. Our main aim was to detect if park remodelling triggers a population decline in these urban green areas. To our knowledge, the effects of these changes to park landscapes on urban biodiversity have never been studied despite being a potential cause of the decline in House Sparrows and other species in urban areas.

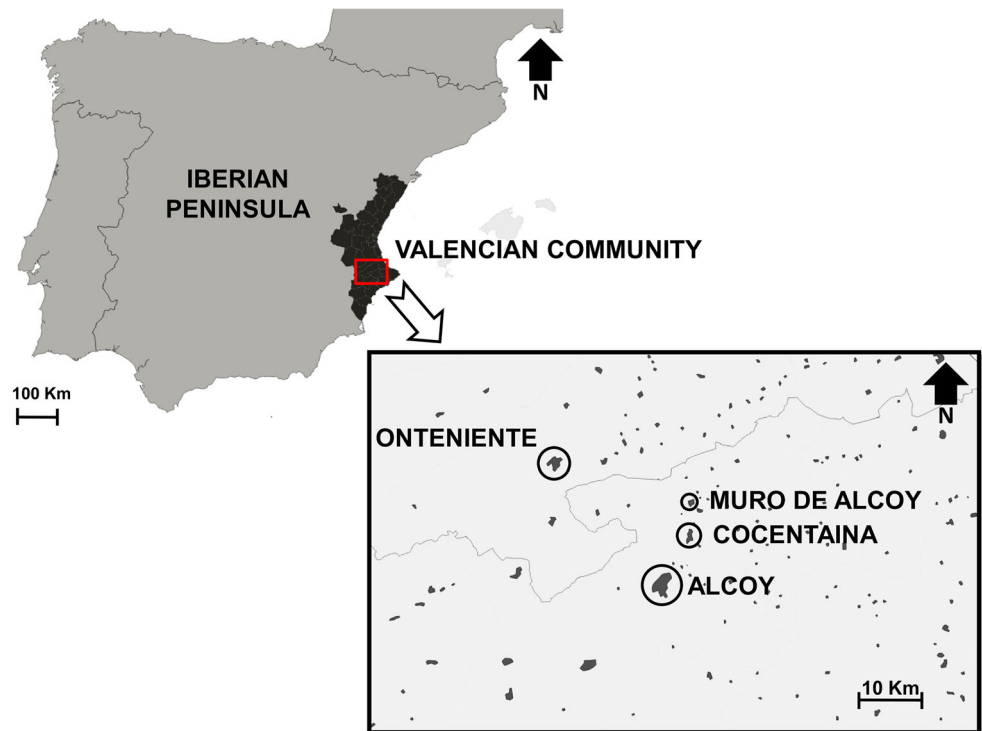
Methods

Study area and bird census

The study was carried out in 32 urban parks distributed into four small to mid-sized towns in adjacent areas in the provinces of Alicante and Valencia (SE Spain; Fig. 1) with between 9000 and 59,000 inhabitants (Table 1). Most of the towns in the region (86%) have fewer than 15,000 inhabitants and 12% are medium-sized (15000–60,000 inhabitants), thus the selected ones are representative of the most frequent type of urban areas in the region (Instituto Nacional de Estadística 2018).

A House Sparrow abundance index was determined in each park as the number of House Sparrows counted during 5 min in one point count of 50 m radius (Gibbons and Gregory 2006) per season in four summers (2015–2018). Counting was done between the end of June and early July, a time that coincides with the end of the breeding season in eastern Spain (Gil-Delgado et al. 1979). The circle around each point count covered an area of 0.79 ha, while the mean size of the 32 parks was 0.90 ha (± 0.01 SE; ranging from 0.01 to 2.80 ha) (Table 2). For the parks whose surface was smaller than 0.79 ha, surrounding streets were included in the surveyed point count. Park areas and distances between them were measured by the SIGPAC cartographic viewer (2018) (Table 2). The mean extension of parks and their shape did not allow us to locate more than one point per park. However given the good detectability of sparrows a single point count

Fig. 1 Location of the four towns selected to study the effects of park remodelling on House Sparrow populations in the Valencian Community (SE Iberian Peninsula) (source: the authors) (QGIS 2018)



per park was considered appropriate to obtain a representative abundance index. Furthermore, for the biggest parks, point counts covered more than the 10% of the park expanse, which is considered a significant part of the study area (Buckland et al. 2015).

Previous studies in the region have shown that although House Sparrows rarely nest in urban parks, these areas are key places for feeding all year round (Bernat-Ponce et al. 2018; Murgui 2009). Thus, by doing counts at the end of the breeding season we expect to include the abundance of the adults breeding nearby and of recently independent young birds. At each point count abundance of birds was recorded by a single observer, to avoid observer bias, for five minutes with a one-minute settlement period. Special attention was paid to avoid double counting and birds were recorded by

sight and vocalisations. Counting was done between the daily official sunrise time and two hours later, which comprises the period when House Sparrows are most active and human activity in parks is minimal (Anderson 2006). Counting was not carried out on rainy and/or windy days.

During the study, we detected remodelling works in 10 parks. The exact dates of remodelling works were provided by the Environment Departments of the studied localities. Remodelling works were mainly carried out in autumn or winter (Table 2); hence the abundance indexes do not reflect their immediate impact but the level of the population some months later. We recorded the type of any important remodelling event that could affect the species in each park, and in particular the type of replaced substrate (natural grass or bare soil) and the type of the new substrate, which included

Table 1 The geographical, demographical and climate characteristics of the towns selected for the study. This table was produced by the authors using data from AVAMET (2018) and the Instituto Nacional de Estadística (2018)

Locality	Coordinates	Number of inhabitants (2017)	Temperature °C (2017)			Rainfall mm (2017)	Altitude (m.a.s.l.)
			Max.	Med.	Min.		
Muro de Alcoy	38°46'47'' N 0°26'10'' O	9319	40.2	16.3	−5.9	458.6	410
Cocentaina	38°44'42'' N 0°26'26'' O	11,461	40.0	16.6	−3.6	390.1	430
Alcoy	38°41'54'' N 0°28'25'' O	59,106	40.4	16.1	−4.4	439.2	562
Onteniente	38°49'20'' N 0°36'26'' O	35,342	38.4	17.2	−3.4	373.0	353

Table 2 Total number of studied parks in each town, mean distance between the studied parks in the same town, mean area (ha) of parks, number of parks remodelled before each census year (number in brackets), remodelling types, the percentage of remodelled surface (in brackets) and remodelling month (in brackets) in these parks. The last row shows the summary of each column, except for the last one

Location	Studied parks	Mean distance (m) between parks $\pm$ SE	Mean area of parks (ha $\pm$ SE)	Parks remodelled before census year	Number, type of modifications, % of remodelled surface and remodelling month
Muro de Alcoy	3	507 $\pm$ 131	0.56 $\pm$ 0.03	—	—
Cocentaina	5	303 $\pm$ 16	0.55 $\pm$ 0.2	2018 (2)	(1) Building an area for dogs on a bare soil area (7.18) (Jan 2018) (1) Brick paving in a bare soil area (10.55) (Jan 2018)
Alcoy	15	385 $\pm$ 41	2.07 $\pm$ 0.7	2016 (2) 2017 (2) 2018 (3)	(1) Concrete paving on natural grass and in an area with trees (16.98) (Feb 2018) (1) Building an area for dogs and replacing bare soil with artificial grass (32.03) (Nov 2015) (1) Replacing natural grass with artificial grass in the whole park (72.03) (Feb-Apr 2018) (1) Concrete and rubber paving in a bare soil area (7.47) (Nov 2016) (3) Replacing natural grass with artificial grass in some areas (1.61/1.42/4.49) (Feb 2018 / Sep-Dec 2015 / Sep 2016)
Onteniente	9	338 $\pm$ 24	0.62 $\pm$ 0.3	2018 (1)	(1) Concrete paving in a bare soil area and installing artificial grass patches (6.20) (Mar 2018)
Total	32	370 $\pm$ 24	0.90 $\pm$ 0.13	2016 (2) 2017 (2) 2018 (6)	

artificial grass, concrete or rubber playground surfaces, building areas for dogs and paving. Remodelled surfaces were measured by the SIGPAC cartographic viewer (2018) (Table 2) and the percentage of remodelled area within each park was calculated. In each studied remodelled park the replaced substrate was of just one type (bare soil or natural grass). The average area of the remodelled (1.21 ± 0.34 SE ha; $n = 10$) and non-remodelled parks (0.76 ± 0.11 SE ha; $n = 22$) were similar ($t = 92.00$; $p = 0.483$).

Statistical analysis

First, we tested if House sparrows' trends and abundance index were similar in remodelled and not remodelled parks before remodelation works. Trends were evaluated by calculating the instantaneous rate of increase (r) (Begon et al. 2006) for each park between consecutive seasons (2015–2016, 2016–2017, 2017–2018) as $\ln(N_{t+1}/N_t)$. The zero abundance values in the counts were replaced with one because it was necessary to calculate natural logarithms to estimate the instantaneous growth rates. This implied that if no sparrow was detected, at least one was considered to be present, which seemed a reasonable assumption given the species' ubiquity. Mean growth rates and abundance of remodelled parks (from the mean of the years before changes in each park) and non-remodelled parks (from

the mean of the four study years in each park) were compared using two sample t -tests. This was useful to check if remodelled parks were a biased sample of the available parks of the study area. These analyses were performed using the Shapiro Wilk normality test ("shapiro.test" function of the "stats" package) (R Core Team 2013) to check for the normality, the Levene test ("leveneTest" of the "car" package) (Fox and Weisberg 2019) to check for homogeneity of variances of both groups of parks (Samuels et al. 2012) and the "t.test" function to perform the two sample t -test ("stats" package). We performed all the statistical analyses in RStudio 3.5.1 (RStudio Team 2018).

Given that sparrows' abundance differed between parks, a relative index of abundance was calculated to compare the average trend shape between the park groups defined by remodelling year or lack of remodelling works. To this end, the abundance index of each park in each year was divided by the maximum recorded abundance index of that park in all four years. In this way, it was possible to evaluate if the parks remodelled in different years presented distinct abundance trends.

We used General Linear Mixed Models (GLMM) to evaluate the park variables that could be related to the changes in the abundance index of House Sparrows. Fixed factors tested in the models were: year of study (2015, 2016, 2017, or 2018 as categories), park remodelling (0: absence; 1: presence; i.e. parks were assigned 0 before remodelling works and 1

afterwards), park size category defined according to thresholds of first and last quartiles of park area distribution (small: < 3685 ha; medium: 3686–12,400 ha; large: > 12,400 ha) and percentage of park area remodelled (0 before remodelling year, constant value after that). Locality and park identity (ID), nested within the former, were considered as random factors. The function “glmmTMB” of the R package “glmmTMB” (Brooks et al. 2017) was used to fit the GLMM.

First of all, we started fitting a GLMM with just the random effects, park identity nested within locality, and using a Poisson distribution. We detected overdispersion after checking this model with the function “check_overdispersion” of the package “performance” (Lüdtke et al. 2019). To avoid this problem, first we fitted to variants of the same model but using the Negative Binomial distribution type 1 (family *nbinom1*) and type 2 (family *nbinom2*) (Blasco-Moreno et al. 2019). In both cases we did not detect overdispersion in the fitted models. The model using the “*nbinom1*” family had lower AIC value, and was selected as a basis to develop subsequent models. Then, we evaluated if random effects could be simplified by examining their variances.

The next step was to include by turn each fixed effect in the GLMM using random intercept and we selected the model that had the lowest AIC. In addition, we tested the significance of fixed effects using the “Anova” function of the “car” package (Fox and Weisberg 2019). Finally, we added the other fixed factors by turn to the best model to check if AIC was reduced by including an additional parameter. For the best model we calculated the intraclass correlation (ICC), to estimate the proportion of the variance in abundance that was accounted for by the random effect, using function “icc” of the package “sjstats” (Lüdtke 2019).

Alteration of lawn or bare soil areas could have different consequences for House Sparrows as the amount of food resources that they provide is likely to be larger in grass-covered areas. Therefore, we checked whether the extension of the remodelling works that affected such types of areas was associated with different declining House Sparrow rates. For this purpose, we used beta regression models, which are appropriate when the dependent variable is a proportion (interval 0, 1) and there is, therefore, neither normally distributed nor homoscedastic (Cribari-Neto and Zeileis 2010). In this analysis, we used the proportion of decline between the year before and the year after remodelling works in the remodelled parks as the dependent variable, and the proportion of remodelled surface and the affected substrate type (lawn $n = 5$; bare soil $n = 5$) as the predictor variables. Beta regression was conducted with the ‘betareg’ function of the ‘betareg’ package (Cribari-Neto and Zeileis 2010).

Results

The mean instantaneous growth rate in the non-remodelled parks ($r = -0.07 \pm 0.06$ SE; $n = 22$) and in the remodelled

parks before the remodelling works took place ($r = -0.09 \pm 0.16$ SE; $n = 8$) were not different (two sample t-test = -0.132 ; $df = 8.76$; $p = 0.898$). Similarly, mean abundance in the non-remodelled parks (13.24 ± 3.05 SE; $n = 22$) and in the remodelled parks before the remodelling works took place (17.12 ± 1.59 SE; $n = 10$) were not different (two sample t-test = 1.127 ; $df = 14.104$; $p = 0.279$). Therefore, on average both the non-remodelled and remodelled parks showed similar population trends and abundance before remodelling of the latter.

The House Sparrows in the non-remodelled parks showed a slightly negative trend, while the remodelled parks presented sharp declines in abundance after the remodelling year (Fig. 2). The pattern was less clear in the parks remodelled between 2016 and 2017 because House Sparrow abundance in one park increased in 2018. Overall, the House Sparrows in the non-remodelled parks declined by about 15% between 2015 and 2018, while the populations in the remodelled parks declined by more than 60% during the same period. When considering only the change in the year interval when a park was remodelled, the abundance index lowered by a mean of 62.33% (± 8.86 SE). Thus, in the remodelled parks, most of the decline between 2015 and 2018 occurred in the year when the parks’ conditions changed.

In the first GLMM with random effects the variance for park identity was 0.255 while variance for locality was almost zero ($1.19\text{e-}09$); hence we ran a new model with just the park identity as random effect which had a lower AIC (Table 3). Then we built random intercept models including one fixed effect each. The best model included the presence of remodelling works with a negative coefficient. Models including the other fixed effects were less plausible as presented $\Delta\text{AIC} > 5$. The addition of further fixed effects led to the model with lowest AIC when the percentage or remodelled area was added, but ΔAIC for the model including just remodelling was 0.2 (Table 3). Regression coefficients both for remodelling and percentage or remodelled area were negative, although the last one did not reach significance ($p = 0.17$). Adding year to this model did not decrease further the AIC. Thus, we have two equally plausible models that include the presence of remodelling works and one of them the percentage of area remodelled. In these models, ICC for the park identity random effect was 0.495 and 0.503, respectively.

In the parks where the replaced substrate was grass, on average the 19.31% ± 13.49 SE ($n = 5$; range 1.42–72.03%) of the surface was remodelled and the 12.69% ± 4.89 SE ($n = 5$; range 6.20–32.03%) in the parks where bare soil was affected. Overall, 16.00% of the parks surface (± 6.85 SE; $n = 10$; range 1.42–72.03%) was affected. The beta regression analysis showed a significant interaction between type of substrate affected and the proportion of park area that was transformed ($z = -2.35$; $p = 0.019$; Appendix Table 4) (Fig. 3), which indicates that the decline proportion changes depending

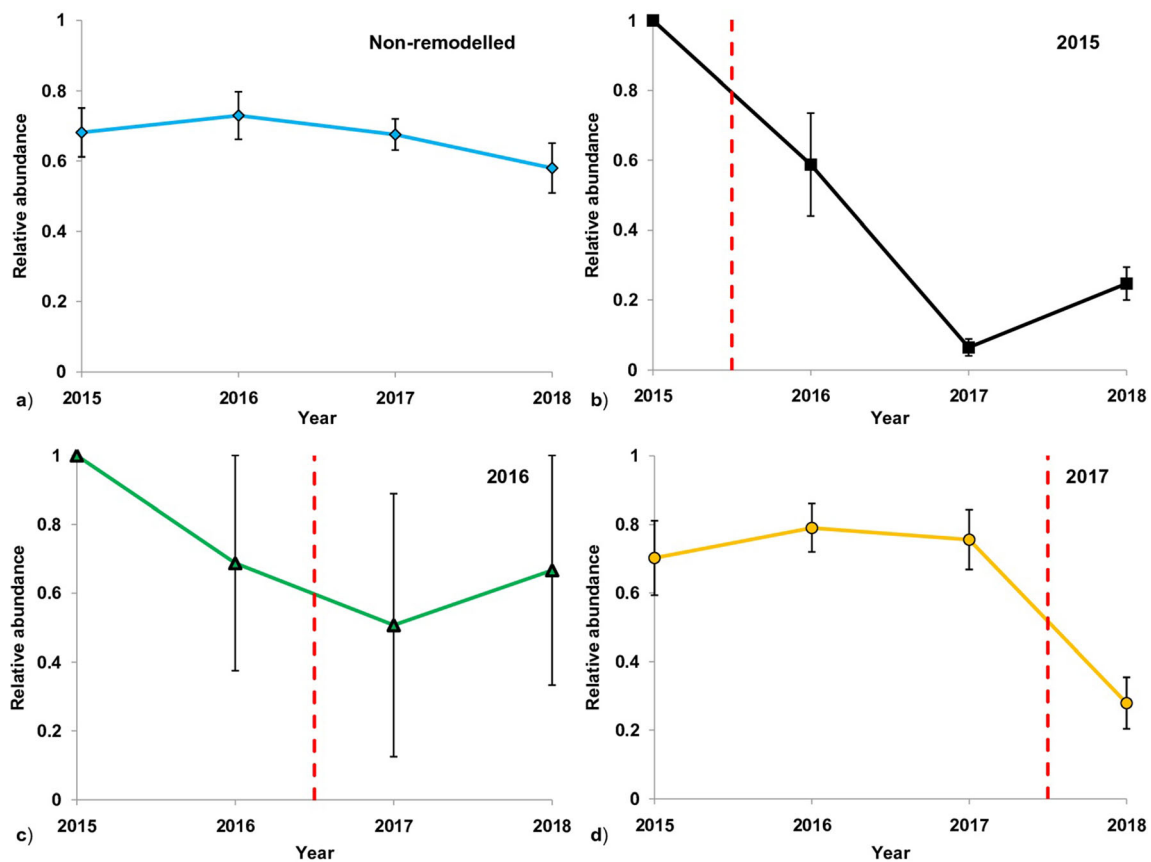


Fig. 2 The House Sparrow relative abundance trends ($\pm$ SE) in the studied parks classified according to the remodelling year. a) Non-remodelled parks ($n = 22$). b) The parks remodelled between the summers of 2015 and 2016 ($n = 2$). c) The parks remodelled between the summers

of 2016 and 2017 ($n = 2$). d) The parks remodelled between the summers of 2017 and 2018 ($n = 6$). The dashed vertical lines mark the remodelling period

on whether the substrate affected by remodelling was lawn or bare soil. Replacement of lawn for abiotic substrates affected more negatively House Sparrow abundance than changing bare soil (Fig. 3).

Discussion

Our results showed that House Sparrow abundance decreased significantly in parks that were remodelled. House Sparrow

Table 3 Fitted GLMM models to evaluate the park variables that could be related to the changes in the abundance index of House Sparrows. Random effects are in brackets. Significant p values are marked with (*)= $p < 0.05$; (**)= $p < 0.01$; (***)= $p < 0.001$. Δ AIC is the difference between the selected model and the model with the lowest AIC value

Model	AIC	Δ AIC
<i>Random effects models</i>		
HS Abund.~ 1 + (1 Locality/Park ID)	868.8	24.8
HS Abund.~ 1 + (1 Park ID)	866.8	22.8
<i>One fixed effects models</i>		
HS Abund.~ Remodelling*** + (1 Park ID)	844.2	0.2
HS Abund.~ % Remodelled area*** + (1 Park ID)	849.4	5.4
HS Abund.~ Year** + (1 Park ID)	859.1	15.1
HS Abund.~ Park area + (1 Park ID)	868.9	24.9
<i>Two fixed effects models</i>		
HS Abund.~ Remodelling*** + % Remodelled area + (1 Park ID)	844.0	0
HS Abund.~ Remodelling*** + Year + (1 Park ID)	845.9	1.9
HS Abund.~ Remodelling** + Park area + (1 Park ID)	846.8	2.8
<i>Three fixed effects models</i>		
HS Abund.~ Remodelling** + Year + Park area + (1 Park ID)	845.6	1.6

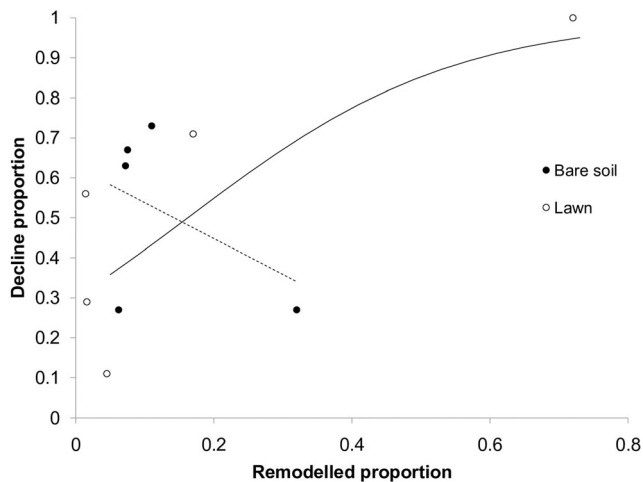


Fig. 3 Beta regression model relating the decline proportion of the House Sparrow abundance index in the first summer after remodelling to the proportion of remodelled area in each park and the affected substrate type (circles: observed data; lines: model prediction: lawn-solid line / bare soil-dashed line)

trends, as measured by the instantaneous growth rate, and abundance index were similar in the parks that were not remodelled and in those that were later remodelled, thus initial differences in the conditions between both groups of parks before works started are unlikely and the most plausible explanation is that changes in park habitat caused by remodelling were responsible for the species decrease. Variability between localities was virtually null, and half of the variance in sparrow's abundance occurred between parks irrespective of their locality, thus the effect of park remodelling was similar in all the studied towns.

Despite 4 years is a short term to precisely determine the effect of park remodelling, abundance lowered immediately after the remodelling was carried out. This scenario indicates a direct link between park remodelling and House Sparrow reductions. This is probably the result of the reduced availability of food resources, which may become even scarcer if suitable foraging areas disappear. In addition, in the few parks modified in 2015 and 2016 abundance tends to remain low several years after remodelling.

It has been suggested that House Sparrow presence and abundance are linked to the available resources offered by green areas (Murgui and Macias 2010; Shaw et al. 2011). Park reforms imply the total or partial replacement of vegetated or soil areas with concrete, asphalt or compact bare ground to build areas for dogs, and even the replacement of natural grass with artificial grass. Even though the park remodelling process is a mixture of diverse changes, they all share the substitution of semi-natural biotic cover (garden plants, weeds and soil) by abiotic artificial substrates (concrete, asphalt, artificial grass), which expectedly reduces food availability for House Sparrows and other urban animal species (Jones and Leather 2012; Smith et al. 2005; Turrini and Knop 2015; Weir

2015). Our results showed that increasing percentage of remodelled area contribute to decrease sparrow abundance, which would be an expected result as it implies a higher reduction in the amount of resources and suitable habitat for the species. In addition, remodelling of the lawn areas of parks is more pernicious for House Sparrow populations than remodelling bare soil areas. Lawn areas are expected to harbour a higher amount of seeds and invertebrates than bare soil, which could explain this result.

To understand the consequences of park remodelling for urban sparrow populations it is necessary to know if the birds that disappeared from remodelled parks died or dispersed to other urban habitats. House Sparrows home ranges in urban habitats vary between 0.11 and 0.86 ha (Cheke 1972; Cramp and Perrins 1994; Heij and Moeliker 1990; Vangestel et al. 2010). By assuming a circular home range, the radius of the wider home range reported would be slightly longer than 50 m. The mean distance between parks in the study area was 370 m, thus House Sparrows may have difficulties in finding new parks after their home park has been remodelled. This suggests that most of the birds that disappeared from the remodelled parks either died or switched foraging areas to adjacent urban habitats, such as streets, which are less suitable zones for House Sparrows (Bernat-Ponce et al. 2018). Despite this, it is possible that some of the sparrows that moved from the remodelled parks reached alternative ones or streets with the presence of hedges, green walls and private gardens (Dover 2015; Moudrá et al. 2018), but they would have to compete with resident sparrows there. The population stability that we found in the non-remodelled parks suggests that density-dependent regulation would prevent abundance from increasing in these parks. Consequently, park remodelling could contribute to a general decline of House Sparrow abundance in cities and more research is needed to evaluate its effect on House Sparrow abundance on the whole town scale.

Valencian towns and cities, like many other European ones, are being reurbanized (Boira 1995; López-Gay 2011, 2014) and their parks are being modified by the process that we name “park remodelling”. The substitution of semi-natural biotic cover by abiotic substrates that we describe in this paper contributes to save water for plant irrigation, and reduces long-term maintenance costs. Thus, it is likely that this process will continue to expand, especially in dry or Mediterranean areas where water availability for irrigation is frequently limited (Yaghoobian et al. 2010). Initially, artificial turf was used to replace grass in sport fields in the 1960s, but nowadays it is spreading to schools, child-care facilities, parks, streets, play areas, home lawns and commercial areas (New Yorkers for Parks 2006; Toronto Public Health 2015). Environmental and health concern about artificial turf areas is growing (Cheng et al. 2014; Claudio 2008); especially in relation to its effects on the urban heat island, carbon and water cycle, and environmental impacts of its rubber components that

derive from scrap tyres (Claudio 2008; Toronto Public Health 2015; Yaghoobian et al. 2010). To our knowledge, no studies have been conducted about its effect on biodiversity, despite the potential ecological consequences that artificial turf has for urban wildlife (Francis 2018). Our results show, for the first time, that replacing semi-natural substrates such as grass or bare soil with artificial grass and other abiotic substrates leads to House Sparrow declines in parks, which could also happen with other species.

Other important feature of the new substrates in the studied urban parks is that they increase their impervious surface. This seems to be part of a general trend for increasing impervious substrates in urban green areas, which has been shown to have an effect on urban hydrological cycle (Warhurst et al. 2014; Perry and Nawaz 2008; Verbeeck et al. 2011). Given this situation, new projects for urban green areas remodelling, either public or private, should consider measures to help maintain them as a suitable habitat for both urban biodiversity and House Sparrows. We suggest prioritizing the traditional design and management of urban green areas and their heterogeneity (Mexia et al. 2018). Furthermore, due to the restricted usual dispersal movements of species, increasing urban green cover, such as tree-lined streets or green stepstones, would favour population connections between parks (De Laet and Trappeniers 2019; Murgui 2007). These urban planning measures, along with the use of native trees, bushes and grass species with fewer watering requirements, might prove beneficial to biodiversity, compensate the water saving produced by artificial grass and reduce its heat island effect (Domene et al. 2005; Moore 2009; Yaghoobian et al. 2010). It would also reduce the amount of management requirements needed by municipalities and would increase the number of harboured invertebrates in urban parks, which would directly benefit urban biodiversity and House Sparrows (Narango et al. 2018; Peach et al. 2008; Turrini and Knop 2015; Vincent 2005). Similar urban planning measures could be taken in already remodelled parks, such as increasing native bush density (Wilkinson 2006), using less insecticides in these green urban areas (Summers-Smith 2003) and appropriate management (Heyman et al. 2017). However, going back to traditional park models would probably be a better option to preserve biodiversity.

Our findings could be useful for suggesting conservation and urban planning measures to be applied on large disparate spatial scales where parks are being, or will be remodelled, and where species are sharply declining (Shaw et al. 2008). More research is urgently needed to precisely identify the short-, mid- and long-term effects of park remodelling and the use of artificial grass on urban biodiversity.

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Appendix

Table 4 Coefficients of the beta regression model relating the decline proportion of the House Sparrow abundance index in the first summer after remodelling to the proportion of remodelled area in each park and the affected substrate type. Phi is the precision parameter of the model

	Estimate	SE	z. value	p value
Intercept	−0.740	0.410	−1.804	0.071
Proportion of remodelled park	5.201	1.620	3.210	0.001
Substrate type	1.259	0.675	1.866	0.062
Interaction (proportion:type)	−8.857	3.768	−2.351	0.019
Phi	6.826	2.942	2.320	0.020

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